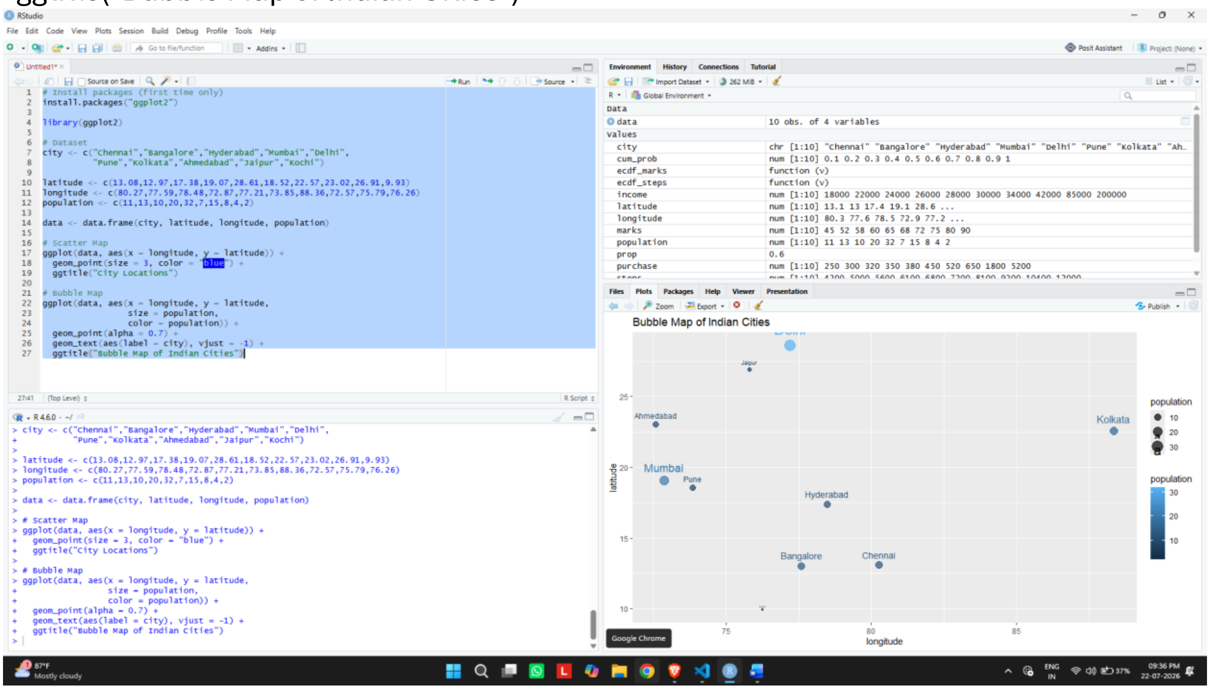
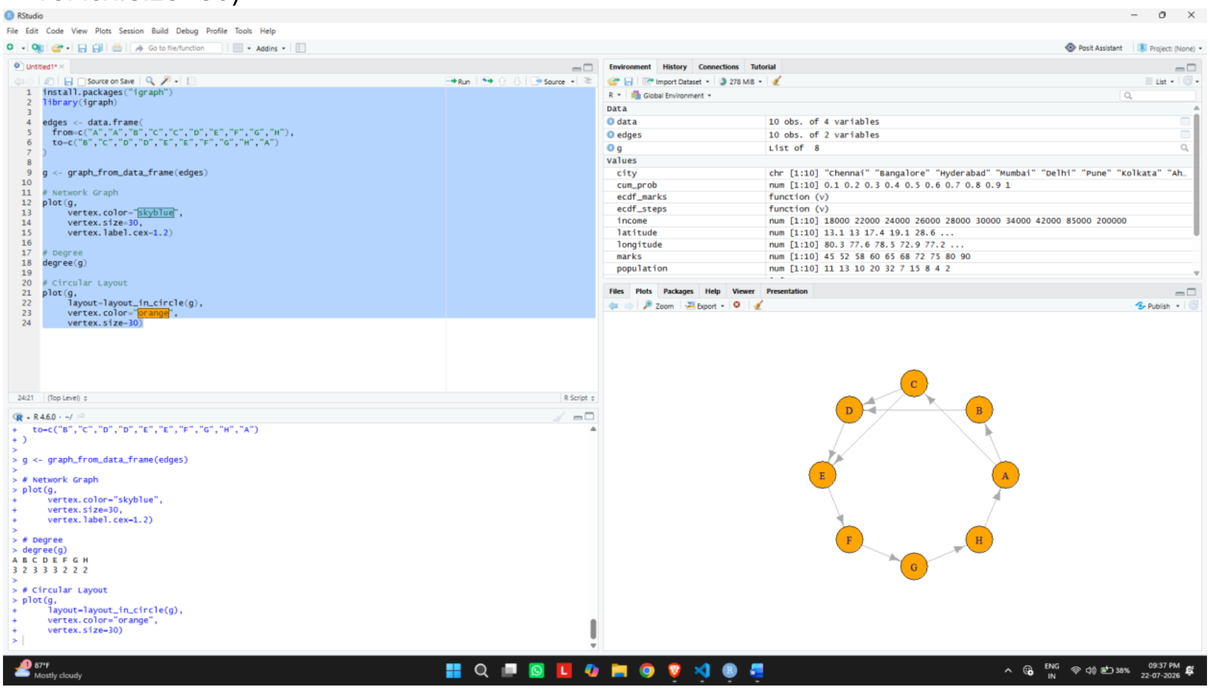
Activity 1 – Geospatial Data Visualization

```
# Install packages (first time only)
install.packages("ggplot2")

library(ggplot2)

# Dataset
city <- c("Chennai","Bangalore","Hyderabad","Mumbai","Delhi",
      "Pune","Kolkata","Ahmedabad","Jaipur","Kochi")

latitude <- c(13.08,12.97,17.38,19.07,28.61,18.52,22.57,23.02,26.91,9.93)
longitude <- c(80.27,77.59,78.48,72.87,77.21,73.85,88.36,72.57,75.79,76.26)
population <- c(11,13,10,20,32,7,15,8,4,2)

data <- data.frame(city, latitude, longitude, population)

# Scatter Map
ggplot(data, aes(x = longitude, y = latitude)) +
 geom_point(size = 3, color = "blue") +
 ggtitle("City Locations")

# Bubble Map
ggplot(data, aes(x = longitude, y = latitude,
          size = population,
          color = population)) +
 geom_point(alpha = 0.7) +
 geom_text(aes(label = city), vjust = -1) +
 ggtitle("Bubble Map of Indian Cities")
```



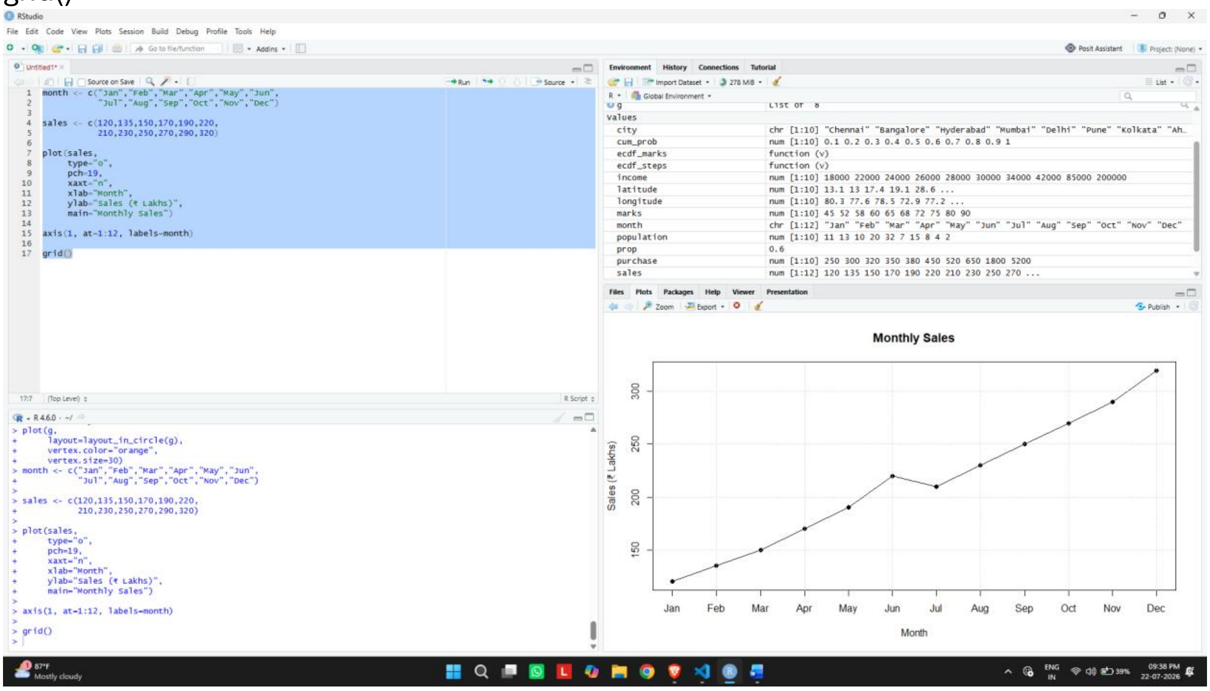
Activity 2 – Network Visualization

```r
install.packages("igraph")
library(igraph)

edges <- data.frame(
from=c("A","A","B","C","C","D","E","F","G","H"),
to=c("B","C","D","D","E","E","F","G","H","A")
)

g <- graph_from_data_frame(edges)

# Network Graph
plot(g,
   vertex.color="skyblue",
   vertex.size=30,
   vertex.label.cex=1.2)

# Degree
degree(g)

# Circular Layout
plot(g,
   layout=layout_in_circle(g),
   vertex.color="orange",
   vertex.size=30)
```



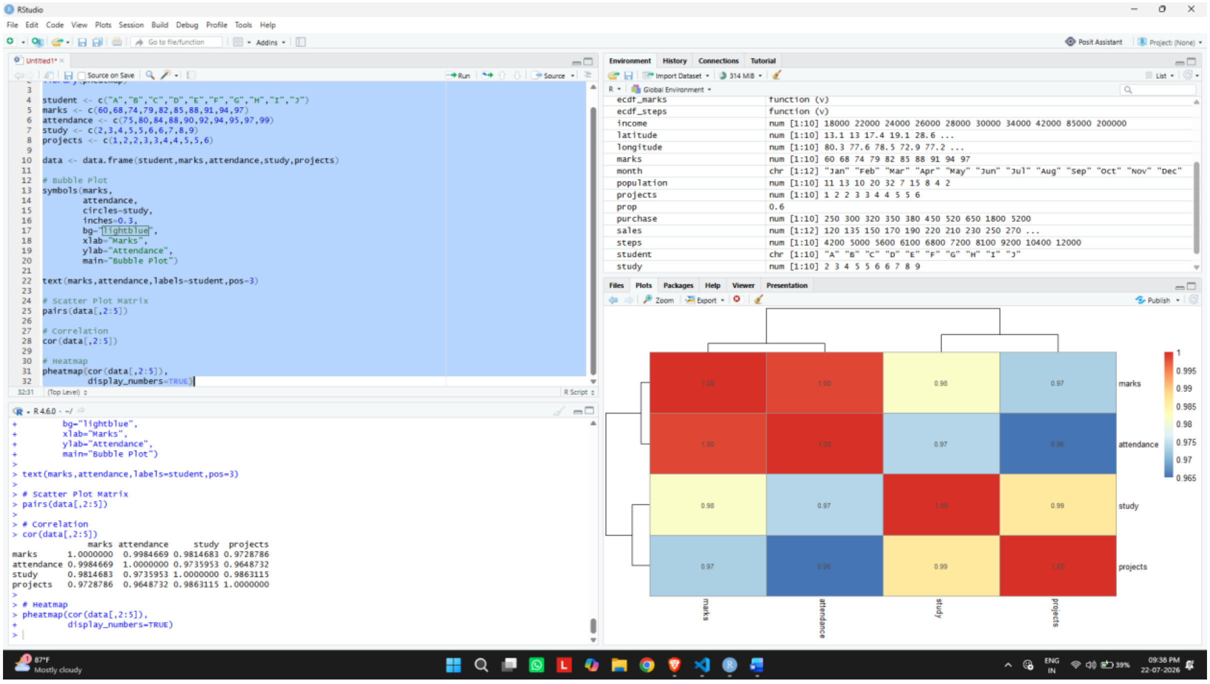
Activity 3 – Temporal Data Representation

```
month <- c("Jan","Feb","Mar","Apr","May","Jun",
      "Jul","Aug","Sep","Oct","Nov","Dec")

sales <- c(120,135,150,170,190,220,
      210,230,250,270,290,320)

plot(sales,
   type="o",
   pch=19,
   xaxt="n",
   xlab="Month",
   ylab="Sales (₹ Lakhs)",
   main="Monthly Sales")

axis(1, at=1:12, labels=month)

grid()
```



Activity 4 – Multivariate Data Visualization

```
install.packages("pheatmap")
library(pheatmap)

student <- c("A","B","C","D","E","F","G","H","I","J")
marks <- c(60,68,74,79,82,85,88,91,94,97)
attendance <- c(75,80,84,88,90,92,94,95,97,99)
study <- c(2,3,4,5,5,6,6,7,8,9)
projects <- c(1,2,2,3,3,4,4,5,5,6)

data <- data.frame(student,marks,attendance,study,projects)

# Bubble Plot
symbols(marks,
    attendance,
    circles=study,
    inches=0.3,
    bg="lightblue",
    xlab="Marks",
    ylab="Attendance",
    main="Bubble Plot")

text(marks,attendance,labels=student,pos=3)

# Scatter Plot Matrix
pairs(data[,2:5])

# Correlation
cor(data[,2:5])

# Heatmap
pheatmap(cor(data[,2:5]),
     display_numbers=TRUE)
```



Activity 5 – Integrated Data Visualization

```
year <- 2016:2025
sales <- c(150,170,200,240,260,300,340,390,430,480)
profit <- c(20,25,32,40,42,55,65,72,82,95)
customers <- c(1200,1500,1800,2200,2400,
           2800,3300,3800,4200,4700)
branches <- c(5,6,7,8,8,10,11,12,13,15)

data <- data.frame(year,sales,profit,customers,branches)

# Line Chart
plot(year,
    sales,
    type="o",
    pch=19,
    col="blue",
    xlab="Year",
    ylab="Sales",
    main="Yearly Sales")

grid()

# Bubble Plot
symbols(sales,
      profit,
      circles=customers,
      inches=0.4,
      bg="orange",
      xlab="Sales",
```

```
    ylab="Profit",
    main="Sales vs Profit")

# Correlation Matrix
cor(data[,2:5])

# Scatter Plot Matrix
pairs(data[,2:5])
```